Practical No: 8

Aim: Installation and Configuration of virtualization using KVM.

Virtualization is software that separates physical infrastructures to create various dedicated resources. It is the fundamental technology that powers cloud computing.

The technology behind virtualization is known as a virtual machine monitor (VMM) or virtual manager, which separates compute environments from the actual physical infrastructure.

Virtualization makes servers, workstations, storage and other systems independent of the physical hardware layer. This is done by installing a Hypervisor on top of the hardware layer, where the systems are then installed.

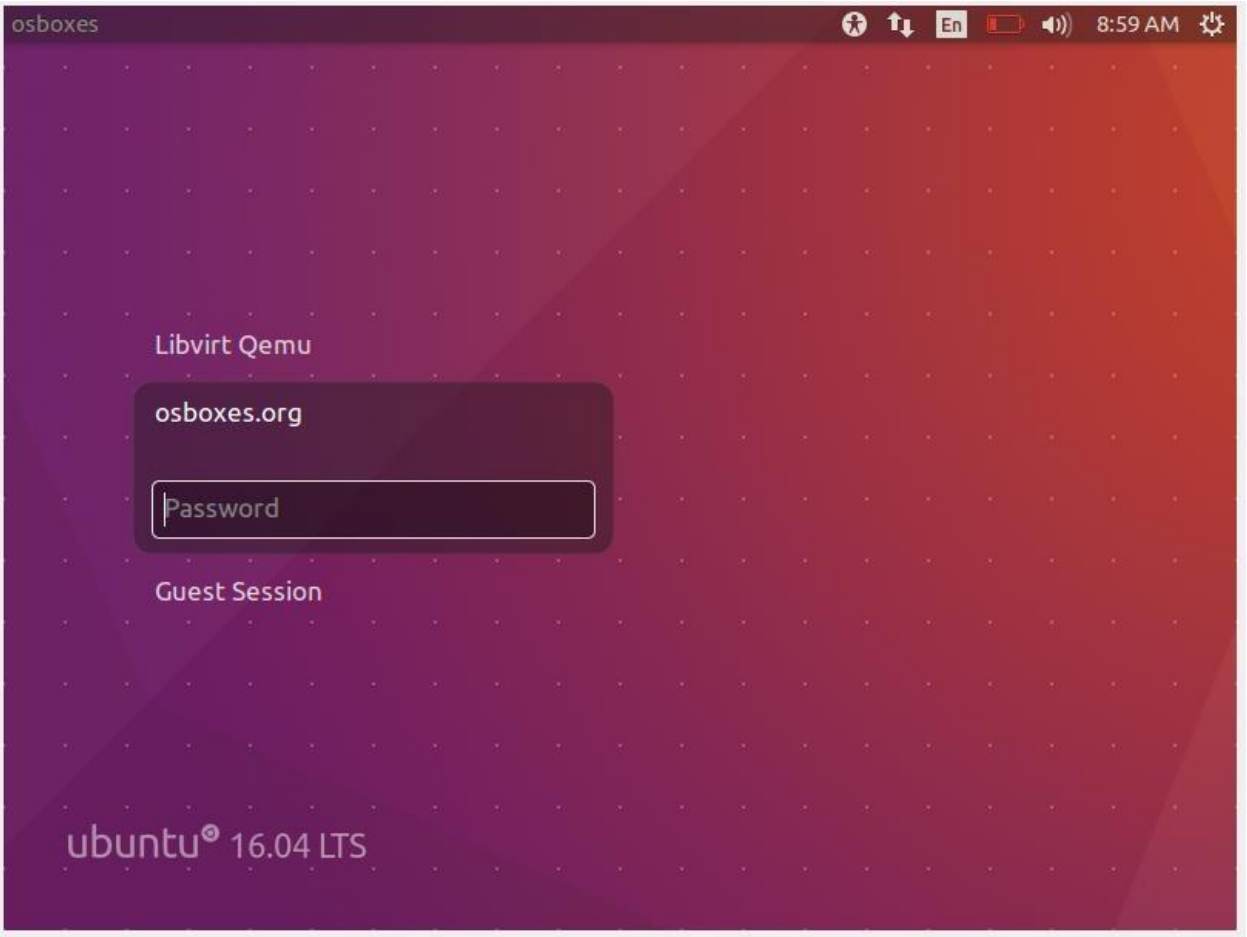
There are three areas of IT where virtualization is making headroads, network virtualization, storage virtualization and server virtualization:

1. Network virtualization
2. Storage virtualization
3. Server virtualization
4. Desktop virtualization

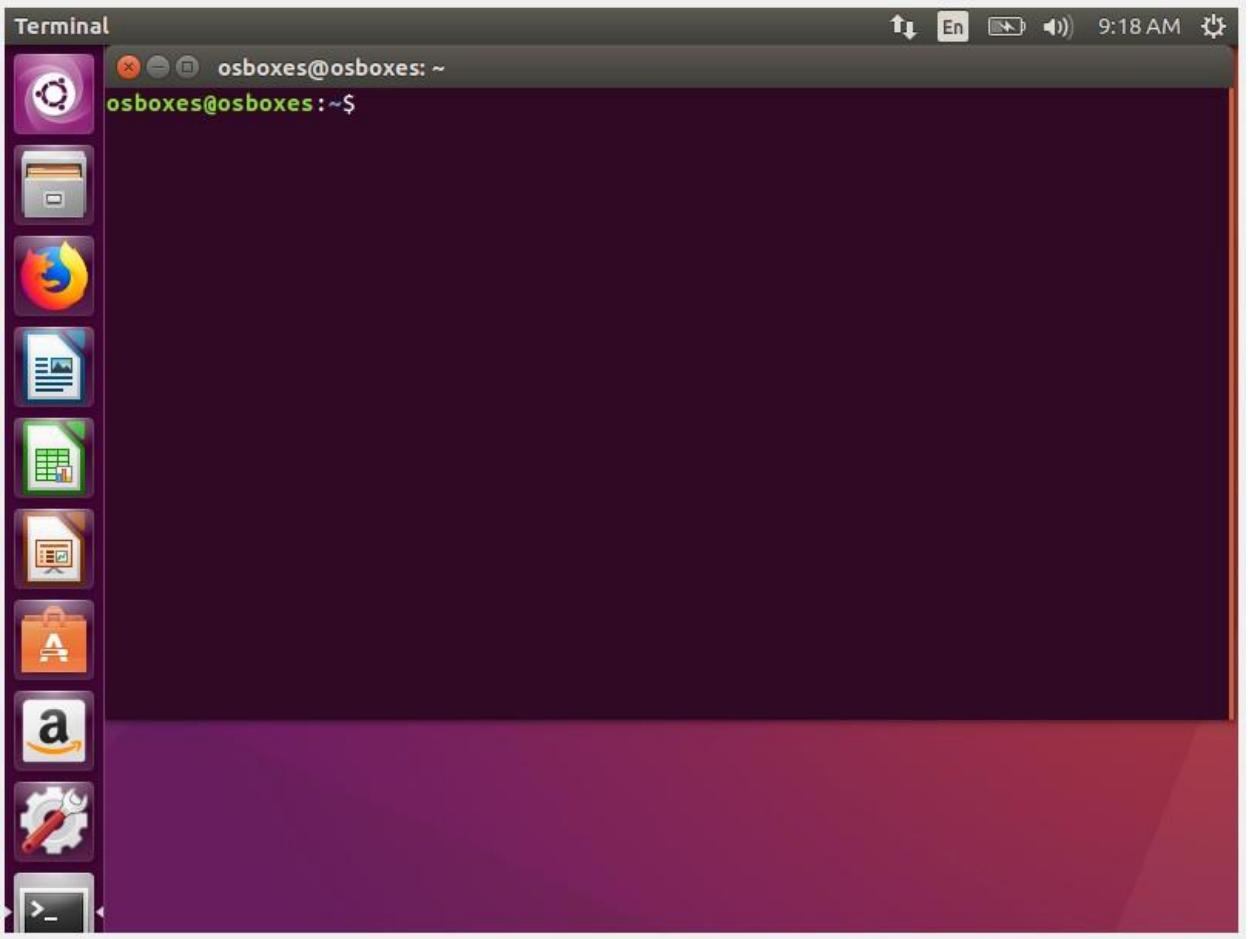
Requirement: Ubuntu operating system, open source software KVM, Internet.

Procedure:

1. Login to Ubuntu OS

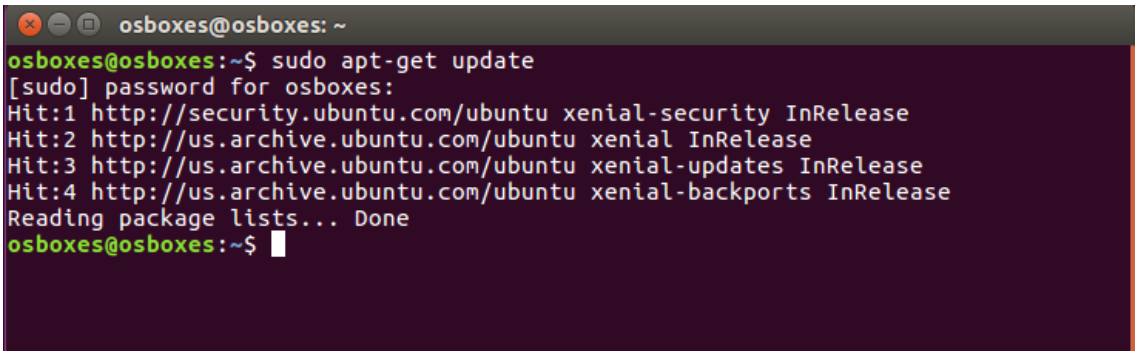


2. Start terminal



3. Install latest update

sudo apt-get update



4. Check whether virtualization enabled on processor

sudo grep -c "svm\|vmx" /proc/cpuinfo

Determine whether svm or vmx CPU extensions are enabled. This command generates output if the CPU is capable of hardware virtualization. However, you may need to enable virtualization in system BIOS for full support.

```
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo apt-get update
[sudo] password for osboxes:
Hit:1 http://security.ubuntu.com/ubuntu xenial-security InRelease
Hit:2 http://us.archive.ubuntu.com/ubuntu xenial InRelease
Hit:3 http://us.archive.ubuntu.com/ubuntu xenial-updates InRelease
Hit:4 http://us.archive.ubuntu.com/ubuntu xenial-backports InRelease
Reading package lists... Done
osboxes@osboxes:~$ sudo grep -c "svm\|vmx" /proc/cpuinfo
0
osboxes@osboxes:~$
```

5. Print the content of /proc/cpuinfo

cat /proc/cpuinfo

```
osboxes@osboxes: ~
osboxes@osboxes:~$ sudo grep -c "svm\|vmx" /proc/cpuinfo
0
osboxes@osboxes:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 142
model name      : Intel(R) Core(TM) i7-7500U CPU @ 2.70GHz
stepping        : 9
cpu MHz         : 2904.000
cache size      : 4096 KB
physical id     : 0
siblings        : 2
core id         : 0
cpu cores       : 2
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
pat pse36 clflush mmx fxsr sse sse2 ht syscall nx rdtscp lm constant_tsc rep_goo
d nopl xtopology nonstop_tsc cpuid pni pclmulqdq ssse3 cx16 pcid sse4_1 sse4_2 x
```

6. install virt-manager

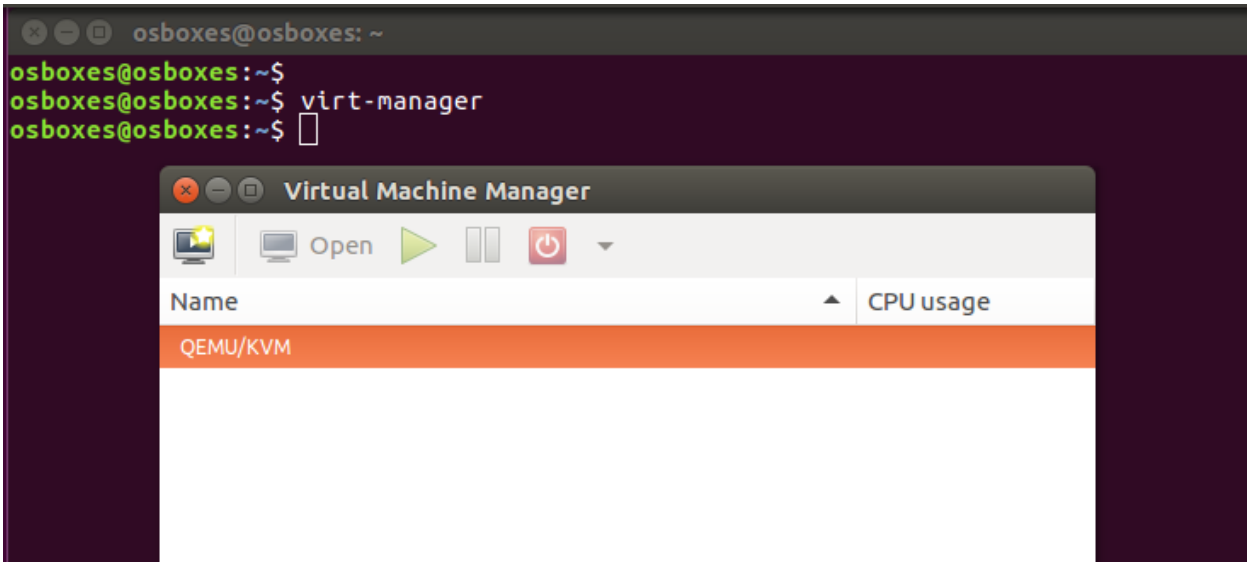
sudo apt-get install qemu-kvm libvirt-bin bridge-utils virt-manager

```
osboxes@osboxes:~$ sudo apt-get install qemu-kvm libvirt-bin bridge-utils virt-m
anager
Reading package lists... Done
Building dependency tree
Reading state information... Done
```

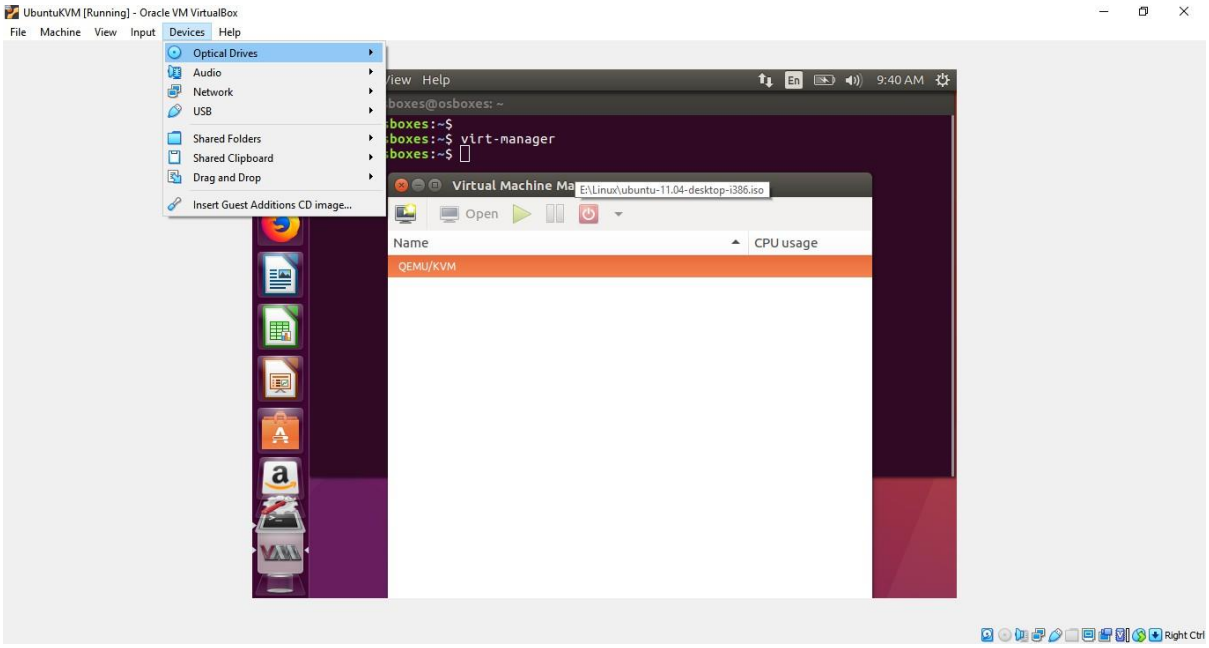
7. Restart the Ubuntu system

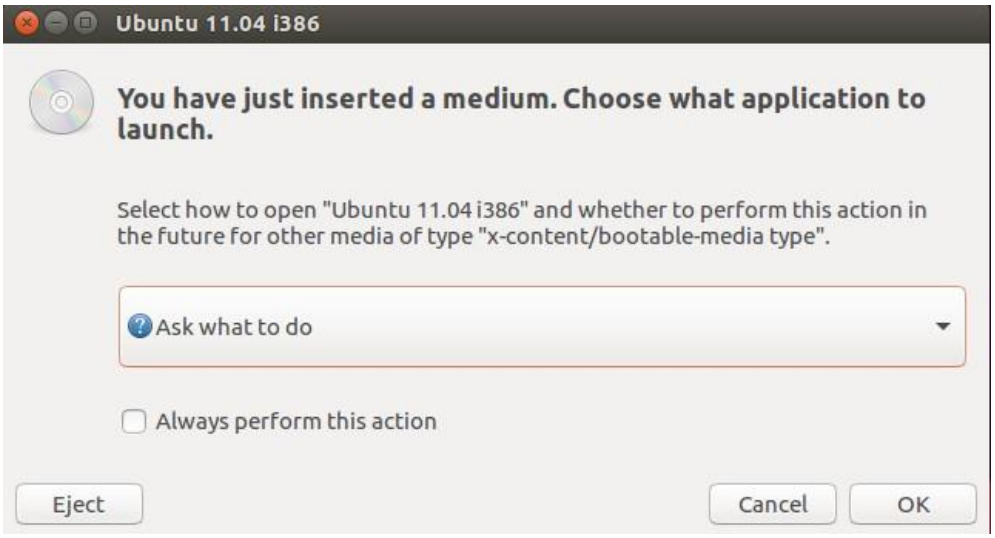
8. Start virt-manager

virt-manager



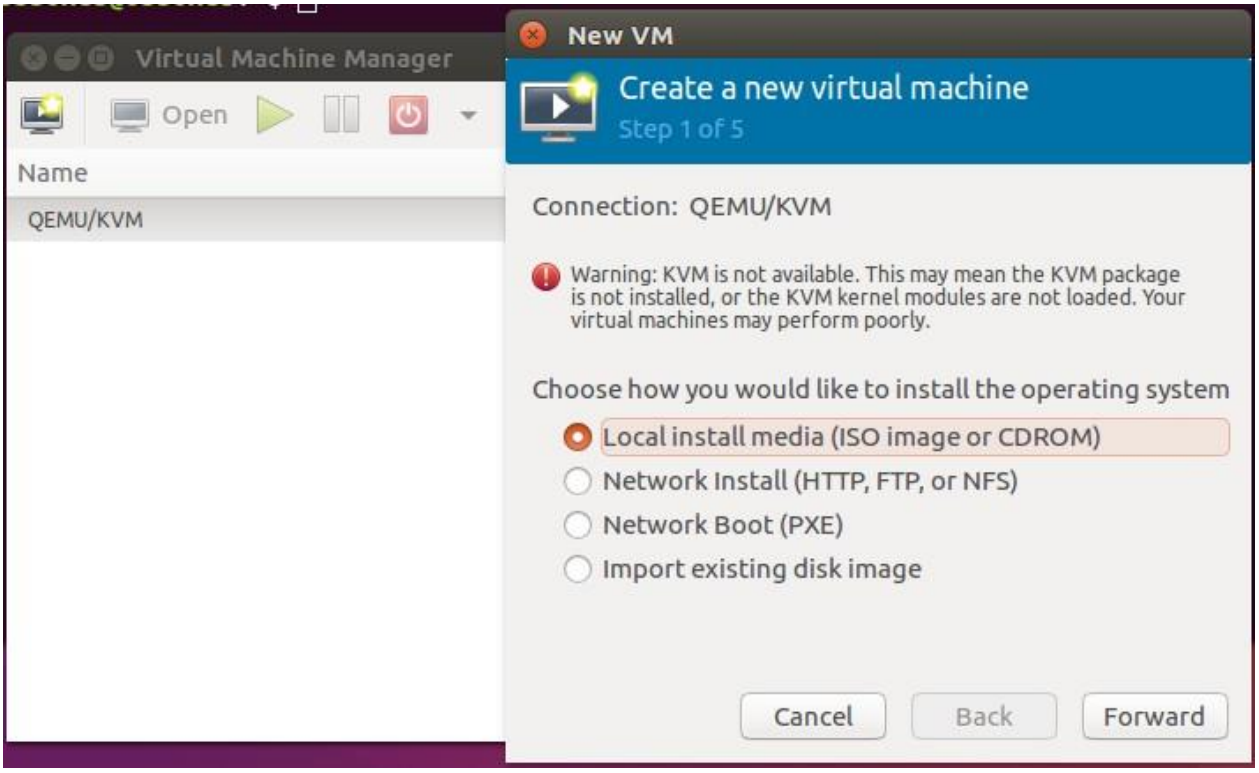
9. Add any OS (.iso) file in Oracle Virtual Box then (.iso) will be loaded in CDROM of Ubuntu VM





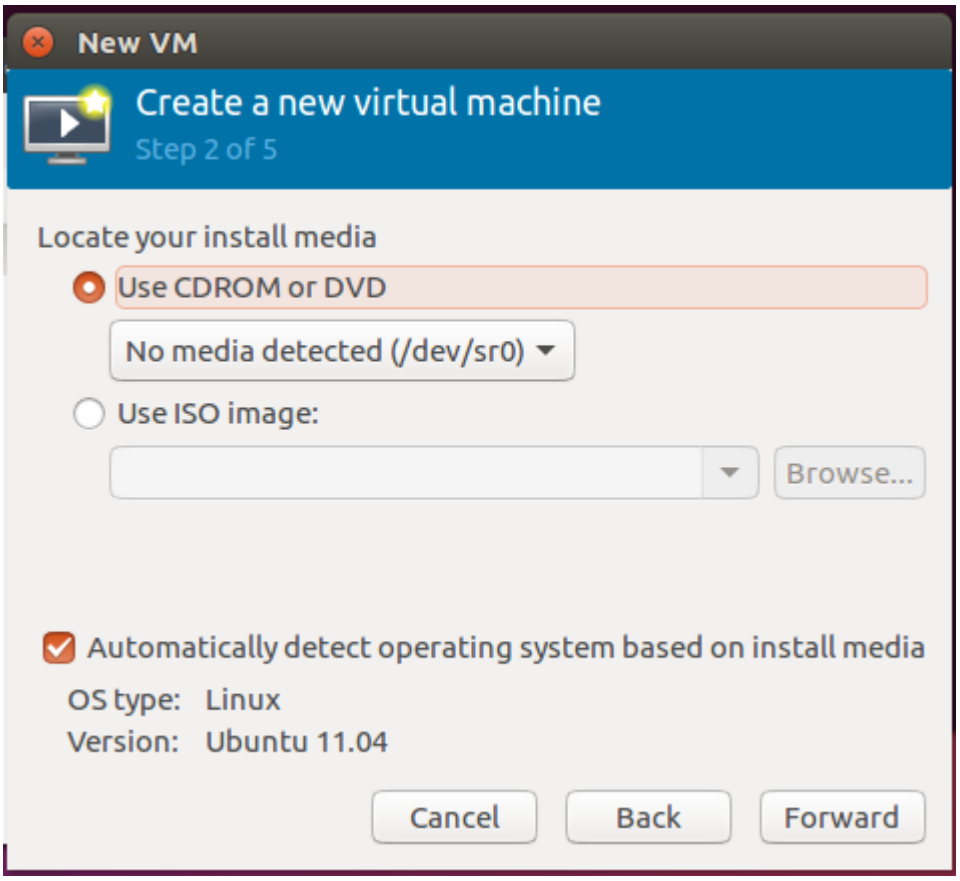
Click on Cancel

10. Click on Create a new virtual machine

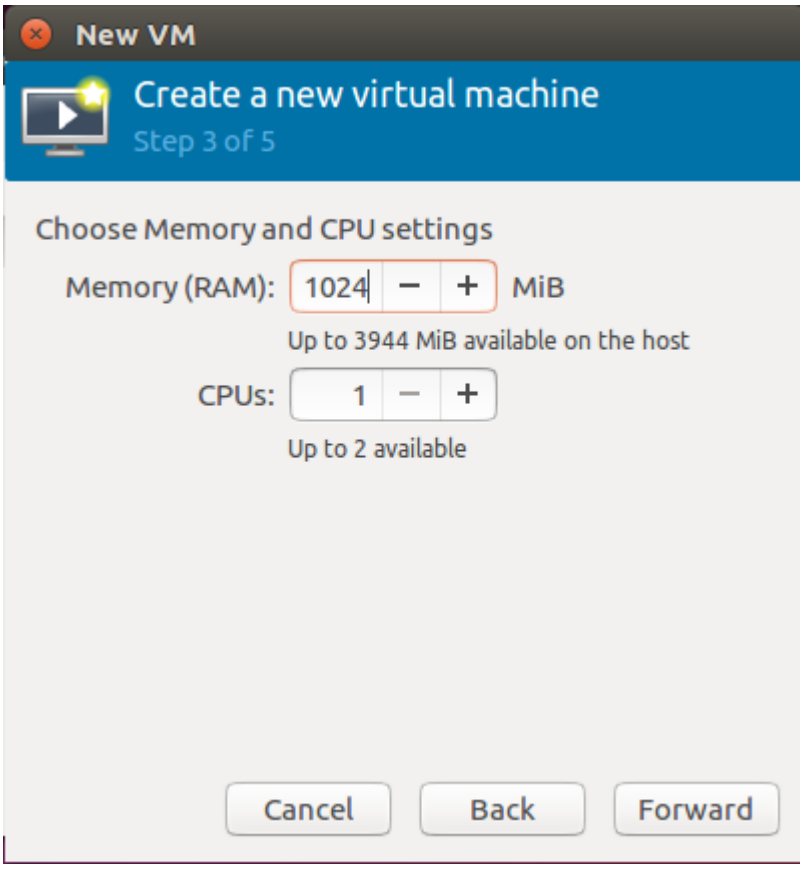


Select Local install media (ISO image or CDROM) and Click on Forward

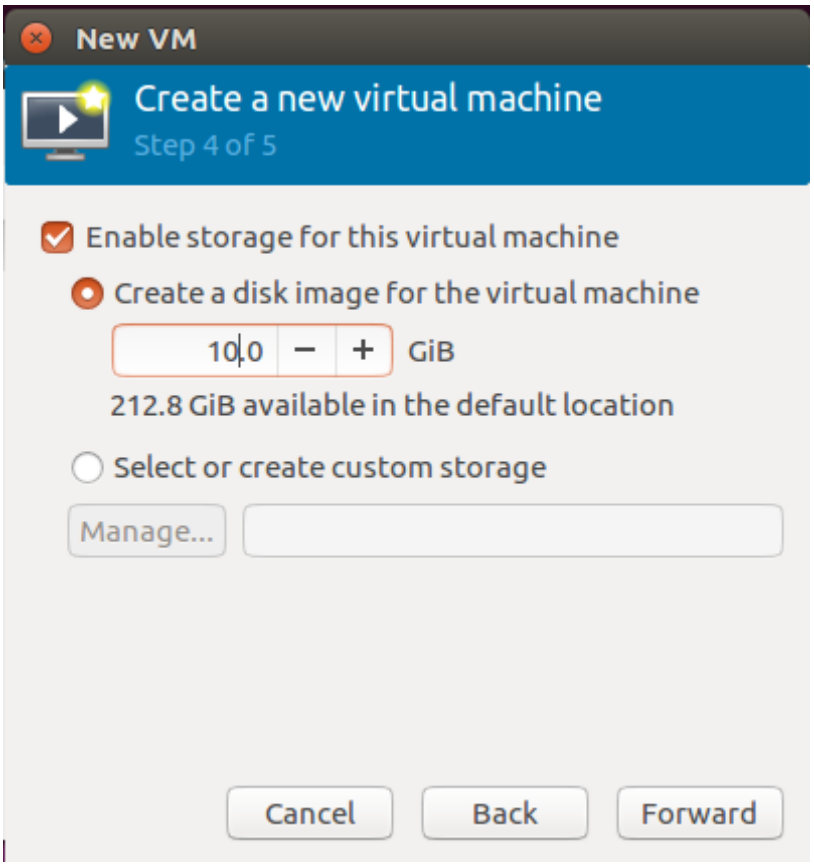
11. Select Use CDROM or DVD and click on Forward



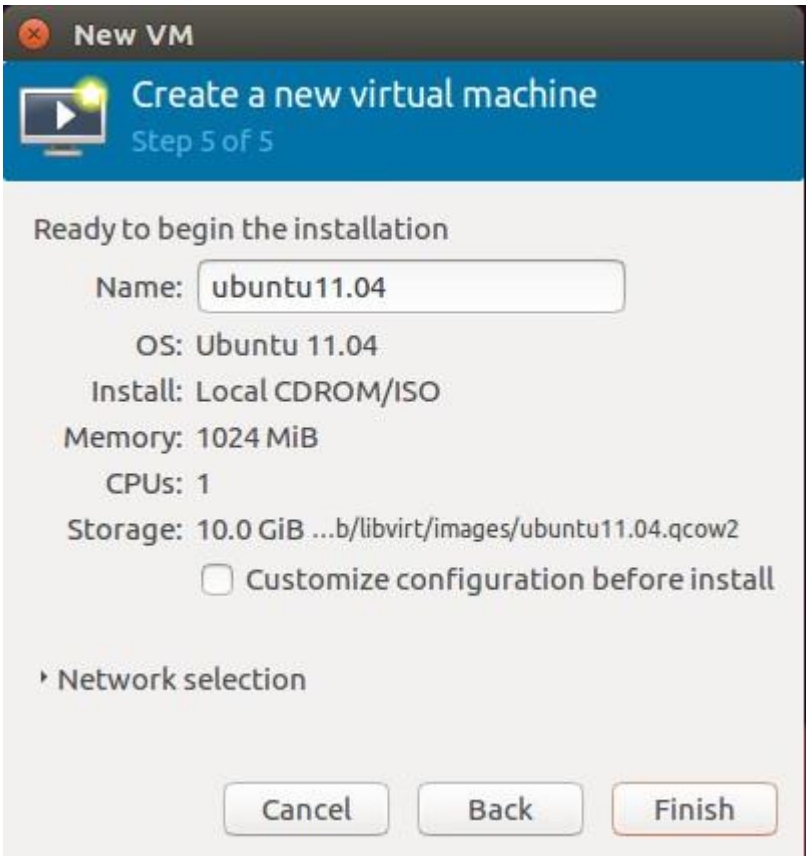
12. Allocate RAM and CPUs and click on Forward



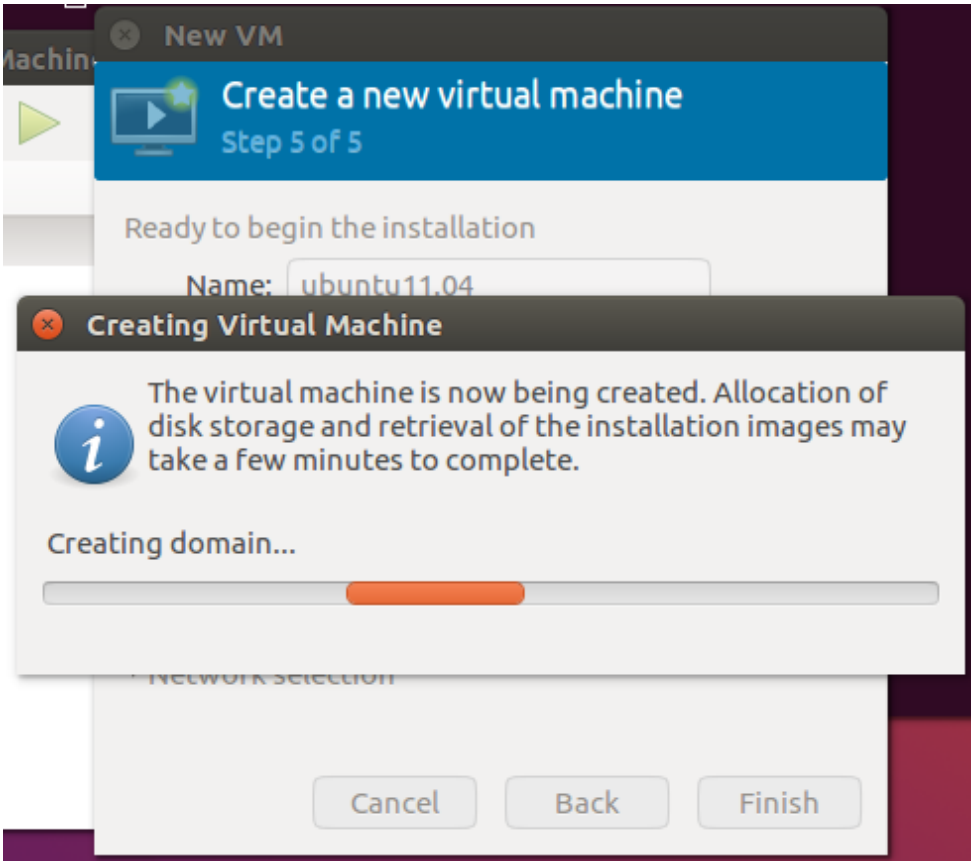
13. Assign Storage space and click on Forward



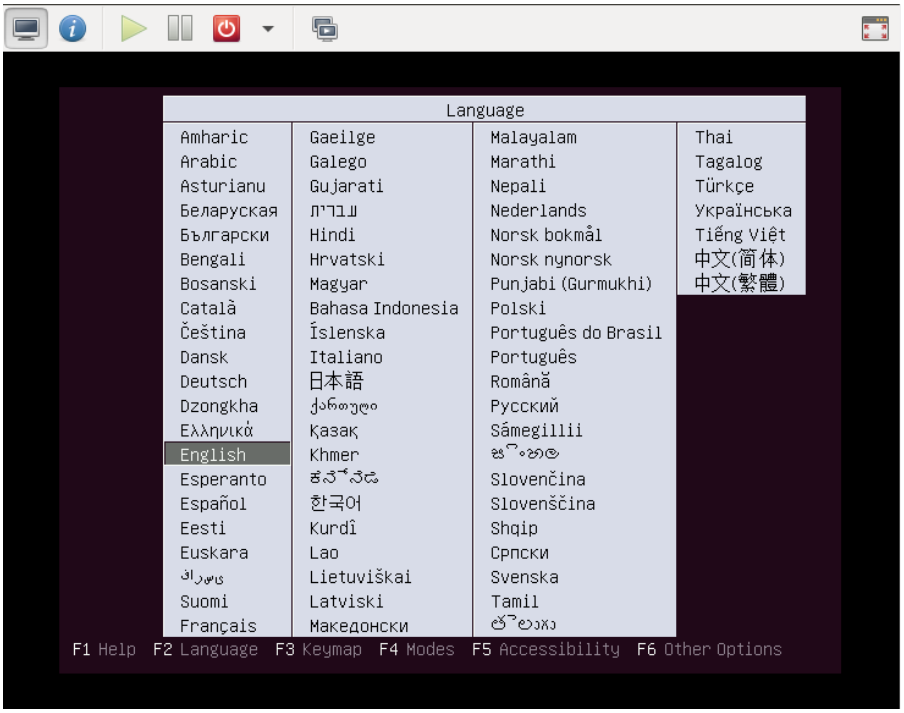
14. Click on Finish

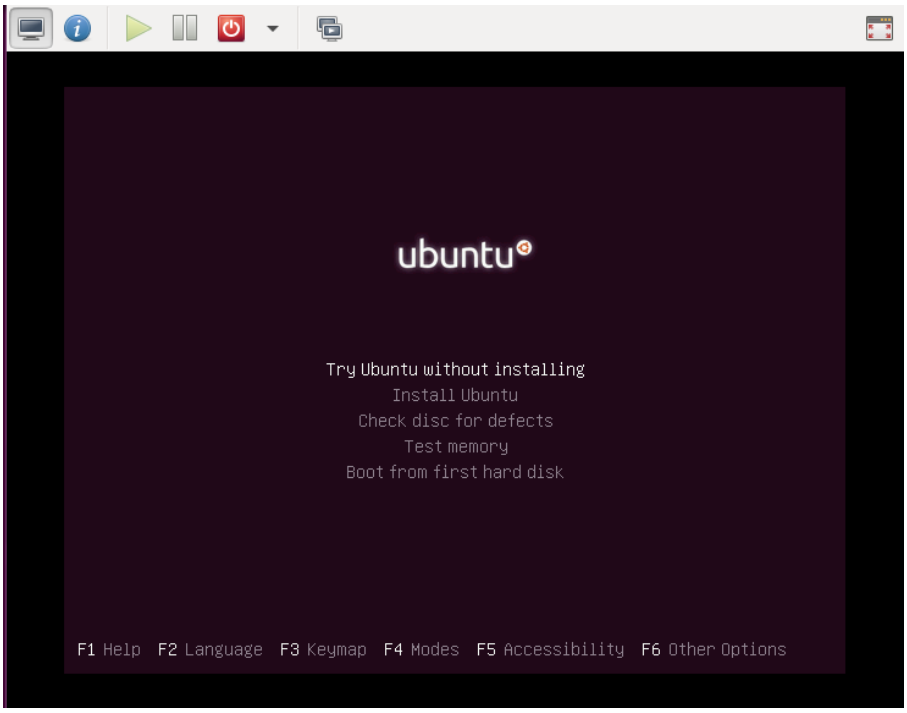


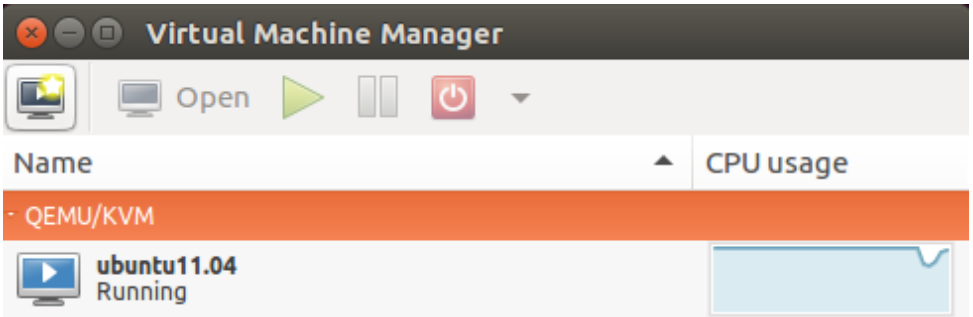
15. Generates .qcow2 file for VM



16. VM will start

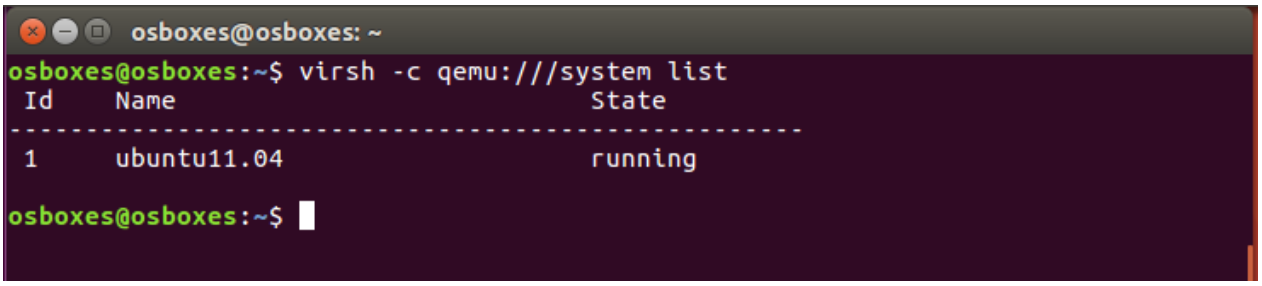
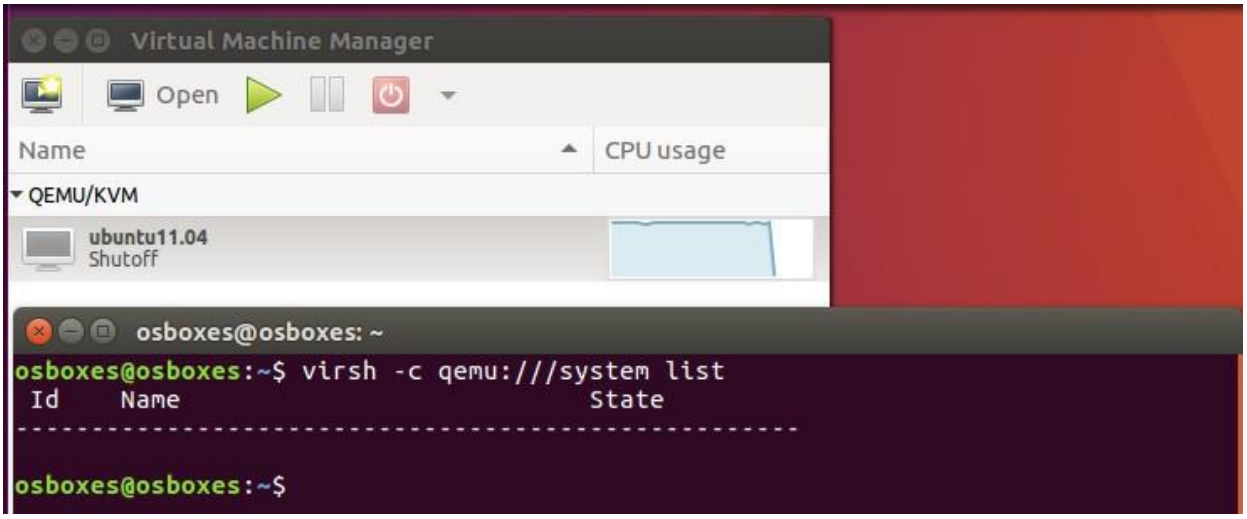






17. We can check VM instances running on terminal

virsh -c qemu:///system list



After shutdown